

# Sinkhorn revisited - joint calibration to European and One-Touch prices

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## Abstract

We apply a Sinkhorn scheme to the problem of constructing an admissible joint law for the terminal value and supremum  $S_T$  of a continuous martingale  $X$ , given target marginals  $\mu$  and  $\nu$ . This corresponds to fitting European and One-Touch prices at all strikes and barrier levels.  $\nu$  necessarily lies between the classical Azéma-Yor and Perkins bounds, but we show this condition is not sufficient with a simple counterexample. Our approach avoids constrained entropy minimization and duality arguments, and after each Sinkhorn cycle, the law of  $S_T$  and the Rogers[Rog93] condition are satisfied exactly, and the  $X_T$ -marginal converges to  $\mu$  (under a finite-entropy feasibility assumption) at rate  $1/\sqrt{n}$  in the total variation metric. As [Rog12] remarks, this problem can also be solved as an optimal transport linear programming problem where we compute the max or min price of a non-tradable option given tradable options, but the latter typically leads to a sparse joint law which may be less realistic from a financial standpoint.

## 1 Background

We first recall the following fundamental result (cf. Theorem 3.1 and Lemma 2.3 in [Rog93]):

**Theorem 1.1** *Fix a  $T > 0$  and  $x_0 \in \mathbb{R}$ . A probability measure  $\pi$  on  $\{(s, x) : -\infty < x \leq s, s \geq x_0\}$  is the joint law of  $(S_T, X_T)$  (where  $S_t = \sup_{0 \leq u \leq t} X_u$ ) for some continuous martingale  $X$  with  $X_0 = x_0$  if and only if  $\int |x| \pi(ds, dx) < \infty$ ,  $\int x \pi(ds, dx) = x_0$ ,  $\lim_{a \rightarrow \infty} a \bar{F}_\nu(a) = 0$  and*

$$\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds \quad (\text{as measures on } (x_0, \infty)), \quad (1)$$

where  $\nu(ds) = \int_{x \leq s} \pi(ds, dx)$ ,  $\eta(ds) = \int_{x \leq s} x \pi(ds, dx)$ , and  $\bar{F}_\nu(s) = \nu((s, \infty))$ .<sup>1</sup>

**Remark 1.1** We give a sketch proof of the sufficiency part of the [Rog93] proof in Appendix D. Eq (1) has the physical interpretation

$$\mathbb{P}(S_T > b)db = \mathbb{E}((S_T - X_T)1_{S_T \in db}) = \mathbb{E}((b - X_T)1_{S_T \in db}). \quad (2)$$

**Remark 1.2 (The integrated form of the Rogers condition)** In [Rog12], the characterisation in Theorem 1.1 is stated in the integrated form: for  $\pi$  supported on  $\{x \leq s, s \geq x_0\}$  with  $\int |x| \pi(ds, dx) < \infty$ ,

$$\int_{\{s \geq a\}} (x - a) \pi(ds, dx) = 0 \quad \text{for all } a \geq x_0, \quad (3)$$

i.e.  $\mathbb{E}(X_T - a; S_T \geq a) = 0$  for all  $a \geq x_0$ , which is optional stopping at the first hitting time to  $a$ . The integrability condition  $\int |x| \pi(ds, dx) < \infty$  is a standing hypothesis in both formulations: it is the first condition of Theorem 1.1, and it is what makes  $\eta$  and the integral in (3) well defined. Under it, (3), which in terms of  $\eta$  and  $\nu$  reads  $\eta([a, \infty)) = a\nu([a, \infty))$  for all  $a \geq x_0$  is equivalent to the remaining three conditions of Theorem 1.1, namely  $\int x \pi(ds, dx) = x_0$ ,  $\lim_{a \rightarrow \infty} a \bar{F}_\nu(a) = 0$  and (1):

- (i) At  $a = x_0$ , (3) is the mean condition, since  $\pi$  is supported on  $s \geq x_0$ .
- (ii) For  $a > 0$ ,  $0 \leq a \bar{F}_\nu(a) \leq a\nu([a, \infty)) = \eta([a, \infty)) = \int x 1_{\{s \geq a\}} \pi(ds, dx)$ . The integrand is dominated by  $|x|$ , which is  $\pi$ -integrable by the standing hypothesis, and it tends to zero pointwise as  $a \rightarrow \infty$  since  $S_T < \infty$ ; hence  $\lim_{a \rightarrow \infty} a \bar{F}_\nu(a) = 0$  by dominated convergence.
- (iii) For (1), we differentiate  $\eta([a, \infty)) = a\nu([a, \infty))$  in  $a$  in the sense of measures to get:

$$-\eta(da) = d_a[a\nu([a, \infty))] = \nu([a, \infty))da - a\nu(da) = \bar{F}_\nu(a)da - a\nu(da) \quad \text{on } (x_0, \infty),$$

which is (1). Conversely, integrating (1) over  $[a, \infty)$  recovers (3) for  $a > x_0$ , and the tail condition makes the boundary term at infinity vanish, and at  $a = x_0$  the mean condition supplies the atom  $\eta(\{x_0\}) = x_0\nu(\{x_0\})$ .

Thus, given integrability, (3) contains the mean and tail conditions automatically, whereas the local form (1) on its own does not, as Example 2.1 shows; it is nevertheless the local form that the Sinkhorn scheme below is built on.

<sup>1</sup>See also Remark (i) after Theorem 3.1 in [Rog93]. The results in [Rog93] are formulated for uniformly integrable (UI) martingales on  $[0, \infty)$  and their terminal value  $M_\infty$ . They apply here because a martingale  $X$  on  $[0, T]$ , extended by  $X_t = X_T$  for  $t \geq T$ , is a UI martingale on  $[0, \infty)$  with  $X_\infty = X_T$  and  $\sup_{t \geq 0} X_t = S_T$ , and conversely a UI continuous martingale can be time-changed onto  $[0, T]$  without altering its terminal value or its supremum. Theorem 1.1 is thus a restatement of Theorem 3.1 and Lemma 2.3 in [Rog93], together with Remark (i) there for  $x_0 \neq 0$ .

## 2 Outline of problem

Our aim is to construct a continuous martingale  $X$  with prescribed marginals  $\mu$  and  $\nu$  for  $X_T$  and  $S_T$ . We refer to (1) as the local Rogers condition, and this condition will form the basis of the Sinkhorn scheme below.

Based on Theorem 1.1, we impose the following:

**Assumption 2.1**  $\int |x|\mu(dx) < \infty$ ,  $\int x\mu(dx) = x_0$  and  $\lim_{a \rightarrow \infty} a\bar{F}_\nu(a) = 0$ , and  $\nu$  lies between the Perkins and Azéma-Yor model-independent bounds, i.e.

$$\sup_{K < B} \left\{ \mu([B, \infty)) + \frac{C_\mu(B) - P_\mu(K)}{B - K} \right\} \leq \nu([B, \infty)) \leq \inf_{K < B} \frac{C_\mu(K)}{B - K}$$

<sup>2</sup> for all  $B \geq x_0$ , where  $C_\mu(K) = \int (x - K)^+ \mu(dx)$  and  $P_\mu(K) = \int (K - x)^+ \mu(dx)$  denote the call and put pricing functions.

**Remark 2.1** If we relax the condition that  $X$  is continuous, the upper bound remains unchanged but the Perkins bound is replaced by the trivial lower bound  $\mathbb{P}^{X \sim \mu}(\max(X, X_0) \geq B)$ , and there is an explicit solution to the problem addressed in this paper described at the end of page 459 and page 460 in [Rog93] using a “lost time” excursion theory construction, whereby  $X_T$  and  $S_T$  are linked via the simple co-monotonic coupling, so there is no need for a Sinkhorn scheme in this case.

**Example 2.1 (Brownian motion stopped at a lower barrier)** (see Remark (ii) after Theorem 3.1 of [Rog93]). Let  $B$  be a standard Brownian motion with  $B_0 = 0$ , let  $\tau = \inf\{t \geq 0 : B_t = -1\}$ , and let  $X_t = B_{t \wedge \tau}$ . Then  $X$  is a continuous local martingale with  $X_\infty = -1$  almost surely, and its supremum  $S_\infty = \sup_{t \geq 0} X_t$  has law

$$\nu(ds) = \frac{ds}{(1+s)^2}, \quad s \geq 0, \quad \text{so that} \quad \bar{F}_\nu(s) = \nu((s, \infty)) = \frac{1}{1+s}.$$

The joint law of  $(S_\infty, X_\infty)$  satisfies the local Rogers condition (1) on  $(0, \infty)$ , but  $X$  is not uniformly integrable:  $\mathbb{E}(X_\infty) = -1 \neq 0$  and  $s\bar{F}_\nu(s) \rightarrow 1$  as  $s \rightarrow \infty$ .

**Proof.** For  $s \geq 0$ , the event  $\{S_\infty > s\}$  is the event that  $B$  reaches  $s$  before  $-1$ , which has probability  $1/(1+s)$  (the gambler’s ruin probability, obtained by applying the optional stopping theorem to  $B$  stopped at the first exit from  $(-1, s)$ ); this gives  $\bar{F}_\nu$  and, on differentiating, the density  $\nu(s) = (1+s)^{-2}$ . Since  $X_\infty = -1$  on every path, the measure  $\eta(ds) = \int x\pi(ds, dx)$  is  $\eta(ds) = -\nu(ds)$ . Hence, for  $s > 0$ ,

$$s\nu(s) - \bar{F}_\nu(s) = \frac{s}{(1+s)^2} - \frac{1}{1+s} = \frac{s - (1+s)}{(1+s)^2} = -\frac{1}{(1+s)^2} = -\nu(s)$$

so  $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds$  on  $(0, \infty)$ , which is Eq (1). On the other hand  $\mathbb{E}X_\infty = -1 \neq X_0$ , so  $X$  is not a UI martingale, and correspondingly  $s\bar{F}_\nu(s) = s/(1+s) \rightarrow 1 \neq 0$ , so the tail condition of Theorem 1.1 fails. ■

As alluded to in the abstract, the following counterexample shows that  $\nu$  being a probability measure and lying between the Perkins and Azéma-Yor bounds is not a sufficient condition for an admissible joint law to exist.

**Counterexample 2.1** Let  $\mu \sim U(-1, 1)$ , so  $x_0 = 0$ , and let  $\nu$  be the law on  $(0, 1)$  with tail cdf

$$\bar{F}_\nu(s) = 1 - s - \frac{7}{2}s(1-s)^5, \quad 0 \leq s \leq 1.$$

The Azéma-Yor and Perkins bounds for  $\mu$  are  $1-s$  and  $1-\sqrt{s}$  respectively, and  $1-\sqrt{s} \leq \bar{F}_\nu(s) \leq 1-s$  on  $(0, 1)$ , so  $\nu$  satisfies Assumption 2.1, but there is no continuous martingale  $X$  with  $X_T \sim \mu$  and  $S_T \sim \nu$ .

**Proof.** Suppose such a martingale exists, and let  $\pi$  be the joint law of  $(S_T, X_T)$ . By Theorem 1.1,  $\pi$  satisfies (1):  $\eta(ds) = s\nu(ds) - \bar{F}_\nu(s)ds$  on  $(0, \infty)$ , where  $\eta(ds) = \int_{x \leq s} x\pi(ds, dx)$ . Since  $X_T \geq -1$  almost surely,  $\eta(ds) \geq -\int_{-1 \leq x \leq s} \pi(ds, dx) = -\nu(ds)$ . Substituting the expression for  $\eta$  from Eq (1) into the left-hand side, this reads  $s\nu(ds) - \bar{F}_\nu(s)ds \geq -\nu(ds)$ , i.e.

$$(1+s)\nu(ds) \geq \bar{F}_\nu(s)ds \quad \text{on } (0, 1),$$

so  $(1+s)\nu(s) \geq \bar{F}_\nu(s)$  for Lebesgue almost every  $s \in (0, 1)$ . But this inequality is violated for  $\nu(s) = -\bar{F}'_\nu(s) = 1 + \frac{7}{2}(1-s)^4(1-6s)$  on a neighbourhood of  $s = \frac{1}{3}$ . ■

Motivated by financial applications, we henceforth assume  $\mu$  only has support on the positive real line. If  $X$  is an asset price and interest rates are zero,  $\mu$  can be inferred from European option prices via the Breeden-Litzenberger formula, and we can interpret  $\nu([B, \infty))$  to be the price of a One-Touch option on  $X$  with barrier level  $B > X_0$ .

<sup>2</sup>the bounds are sharp here, and attained by the Perkins and Azéma-Yor Skorokhod embeddings respectively, see the original papers [AY79],[Per86] and [Obl04],[Cox04],[Hob11] for details.

## 2.1 Sinkhorn Integral equations

We henceforth further assume that  $\mu, \nu$  have densities with respect to the Lebesgue measure, and set

$$r(s) = s\nu(s) - \bar{F}_\nu(s)$$

i.e. the density on the right-hand side of (1).

Let  $p(s, x)$  be a candidate joint density for  $(S, X)$ . Based on the  $\mathbb{E}(X_T 1_{S_T \in db})$  term in (2) and the entropic optimal transport literature, we make the exponential ansatz:

$$p(s, x) = R(s, x)e^{u(x)+v(s)+x\lambda(s)}$$

where  $R$  is a reference density (see Assumption 2.2 for conditions on  $R$ ). Then the  $X$ -marginal condition is

$$\mu(x) = e^{u(x)} \int_x^\infty R(s, x)e^{v(s)+\lambda(s)x} ds$$

which we can rearrange as

$$u(x) = \log \mu(x) - \log \left( \int_x^\infty R(s, x)e^{v(s)+\lambda(s)x} ds \right) \quad (4)$$

if the integral term lies in  $(0, \infty)$  (this will be addressed in Lemma C.1). Similarly, the  $S$ -marginal condition is

$$\nu(s) = e^{v(s)} \int_0^s R(s, x)e^{u(x)+\lambda(s)x} dx$$

and the local Rogers condition in (1) is

$$r(s) = e^{v(s)} \int_0^s x R(s, x)e^{u(x)+\lambda(s)x} dx.$$

Dividing by the  $S$ -marginal equation then gives

$$\frac{\int_0^s x R(s, x)e^{u(x)+\lambda(s)x} dx}{\int_0^s R(s, x)e^{u(x)+\lambda(s)x} dx} = b(s) := \frac{r(s)}{\nu(s)} \quad (5)$$

where  $b(s)$  is defined for  $\nu$ -almost every  $s$ . Under suitable technical conditions (see Lemma 2.3 for details),  $\lambda(s)$  can be computed from (5) as a one-dimensional root-finding exercise. The  $S$ -marginal equation gives

$$v(s) = \log \nu(s) - \log \left( \int_0^s R(s, x)e^{u(x)+\lambda(s)x} dx \right). \quad (6)$$

Hence we see that (4), (5) and (6) are the three coupled equations for  $u, \lambda$  and  $v$ .

## 2.2 The Sinkhorn scheme

The coupled equations above suggest the following iterative scheme. Set  $u_0 = v_0 = \lambda_0 = 0$ . Given  $(u_n, v_n, \lambda_n)$ , first set

$$u_{n+1}(x) = \log \mu(x) - \log \left( \int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x} ds \right). \quad (7)$$

This enforces the  $X$ -marginal exactly. Next, choose  $\lambda_{n+1}(s)$  as the solution of

$$\frac{\int_0^s x R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx}{\int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx} = b(s). \quad (8)$$

This is a scalar root-finding problem in  $\lambda$  for each  $s$ ; existence and uniqueness of the root for  $\nu$ -almost every  $s$  are established in Lemma 2.3. Finally set

$$v_{n+1}(s) = \log \nu(s) - \log \left( \int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx \right). \quad (9)$$

The next  $u$ -update restores the  $X$ -marginal, which is the only constraint disturbed by the completed  $(\lambda, v)$  update. In practice, one iteration therefore consists of an explicit  $u$ -update, one scalar root-find for  $\lambda_{n+1}(s)$  for every  $s$ , and an explicit  $v$ -update.

**Definition 2.1** *A probability measure on  $\{(s, x) : -\infty < x \leq s < \infty, s \geq x_0\}$  is called **admissible** if it has  $X$ -marginal  $\mu$ ,  $S$ -marginal  $\nu$ , and satisfies the conditions in Theorem 1.1.*

For probability measures  $P$  and  $Q$  we write  $H(P|Q) = \int \log \frac{dP}{dQ} dP$  if  $P \ll Q$  and  $H(P|Q) = +\infty$  otherwise. The integral is well defined in  $[0, \infty]$ : writing  $\rho = dP/dQ$ ,  $\int (\log \rho)^- dP = \int \rho (\log \rho)^- dQ \leq 1/e$ , since  $\rho \log \rho \geq -1/e$  for  $\rho \geq 0$ , and  $H(P|Q) \geq 0$  by Jensen's inequality.<sup>3</sup>

From here on we make the following further assumption:

**Assumption 2.2**  $\mu = 0$  outside  $0 \leq x \leq K$ ,  $\nu = 0$  outside  $X_0 \leq s \leq K$  for some  $K < \infty$ <sup>4</sup>, and  $\mu(x) > 0$  for Lebesgue almost every  $x \in (0, K)$  and  $\nu(s) > 0$  for Lebesgue almost every  $s \in (X_0, K)$ . The reference density  $R$  is a probability density supported on

$$s \geq X_0, \quad 0 < x \leq s$$

with  $R(s, x) > 0$  for every  $s \geq X_0$  and Lebesgue almost every  $x \in (0, s)$ . Finally, there exists an admissible probability density  $p^*$  such that  $H(p^*|R) < \infty$ .

Since  $H(p^*|R) < \infty$ , we have  $p^* \ll R$ , so  $0 < X \leq S$   $p^*$ -almost surely. Since  $p^*$  is a density,  $p^*(X = S) = 0$ . Hence, using  $b(s) = \mathbb{E}_{p^*}(X|S = s)$ , (by the local Rogers condition), we obtain

$$0 < b(s) < s \tag{10}$$

for  $\nu$ -almost every  $s$ .

**Remark 2.2 (Reference density for Geometric Brownian motion)** A natural choice for the reference density  $R$  is the joint law of  $(S_T, X_T)$  for driftless geometric Brownian motion  $X_t = X_0 e^{\sigma W_t - \sigma^2 t/2}$ . Writing  $Y_t = \log X_t$ , a Brownian motion with drift  $-\sigma^2/2$  and volatility  $\sigma$  started at  $\log X_0$ , the reflection principle and Girsanov's theorem give the joint density of  $M_T = \max_{t \leq T} Y_t$  and  $Y_T$ , and the change of variables  $s = e^b$ ,  $x = e^y$  then yields

$$R(s, x) = \frac{2 \ell(s, x)}{\sigma^3 \sqrt{2\pi T^3} s x} \sqrt{\frac{X_0}{x}} \exp\left(-\frac{\ell(s, x)^2}{2\sigma^2 T} - \frac{\sigma^2 T}{8}\right), \quad \ell(s, x) = \log \frac{s^2}{x X_0},$$

for  $x > 0$  and  $s \geq \max(x, X_0)$ , and  $R(s, x) = 0$  otherwise. Its  $S$ -marginal is

$$\bar{F}_R(B) = \int_B^\infty \int_0^s R(s, x) dx ds = \Phi\left(\frac{-\log(B/X_0) - \sigma^2 T/2}{\sigma \sqrt{T}}\right) + \frac{X_0}{B} \Phi\left(\frac{-\log(B/X_0) + \sigma^2 T/2}{\sigma \sqrt{T}}\right), \quad B \geq X_0,$$

where  $\Phi$  is the standard normal distribution function. Since geometric Brownian motion is a continuous martingale,  $R$  satisfies the local Rogers condition (1) exactly, and it is positive on the interior of its support, as required in Assumption 2.2. Note however that  $R$  is not supported on  $s \leq K$ , so  $q_0 = R$  charges the region  $\{x > K\}$ ; this is why several statements in Section 2 and Appendix C are restricted to  $n \geq 1$ .

**Lemma 2.3** Suppose Assumption 2.2 holds. Let  $u$  be measurable and finite Lebesgue almost everywhere on  $(0, K)$  and suppose that, for  $\nu$ -almost every  $s$ , there exists at least one finite value of  $\lambda$  for which  $0 < \int_0^s R(s, x) e^{u(x) + \lambda x} dx < \infty$ . Then for  $\nu$ -almost every  $s$  there exists a unique finite  $\lambda(s)$  such that

$$\frac{\int_0^s x R(s, x) e^{u(x) + \lambda(s)x} dx}{\int_0^s R(s, x) e^{u(x) + \lambda(s)x} dx} = b(s). \tag{11}$$

Moreover,  $\lambda$  admits a measurable version, and for  $\nu$ -almost every  $s$ , the integral  $\int_0^s R(s, x) e^{u(x) + \lambda x} dx$  is positive and finite for every finite  $\lambda$ , and in particular for  $\lambda = \lambda(s)$ .

**Proof.** See Appendix A. ■

### 3 Convergence for the Sinkhorn scheme

For the convergence argument, define  $p_{i,j,k}(s, x) = R(s, x) e^{u_i(x) + v_j(s) + \lambda_k(s)x}$ . Thus one complete cycle is  $p_{n,n,n} \rightarrow p_{n+1,n,n} \rightarrow p_{n+1,n+1,n} \rightarrow p_{n+1,n+1,n+1}$ , corresponding to the  $u$ -update, the  $\lambda$ -update and the  $v$ -update respectively. By Eq (7),  $(p_{n+1,n,n})^X = \mu$ , so  $p_{n+1,n,n}$  is a probability density. By Eq (9),  $(p_{n+1,n+1,n+1})^S = \nu$ , so every completed iterate is also a probability density. Moreover, Eq (8) gives

$$\frac{\int_0^s x p_{n+1,n+1,n+1}(s, x) dx}{\int_0^s p_{n+1,n+1,n+1}(s, x) dx} = b(s).$$

Since the denominator is  $\nu(s)$  by construction, we see that  $\int_0^s x p_{n+1,n+1,n+1}(s, x) dx = b(s)\nu(s) = r(s)$ . Hence every completed iterate  $p_{n,n,n}$  with  $n \geq 1$  not only has the correct  $S$ -marginal but satisfies the local Rogers condition. The intermediate density  $p_{n+1,n,n+1}$  need not have mass one, so in the entropy calculation below the  $\lambda$ - and  $v$ -updates are treated together.

<sup>3</sup>The same bound justifies computing  $H(P|Q)$  as a double integral computed one variable at a time whenever  $P$  and  $Q$  are given by densities.

<sup>4</sup>note there is mild overloading of notation here; this  $K$  is not the strike price variable  $K$  in Assumption 2.1

**Lemma 3.1** For every  $n \geq 0$  for which the  $u$ -update is well defined,

$$\frac{p_{n+1,n,n}(s, x)}{p_{n,n,n}(s, x)} = \frac{\mu(x)}{(p_{n,n,n})^X(x)} \quad (12)$$

and

$$H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X). \quad (13)$$

**Proof.** Since  $p_{n+1,n,n}$  and  $p_{n,n,n}$  differ only in the  $u$ -potential,  $p_{n+1,n,n}(s, x)/p_{n,n,n}(s, x) = e^{u_{n+1}(x)-u_n(x)}$ . Integrating  $p_{n,n,n}(s, x) = R(s, x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$  over  $s$  gives  $(p_{n,n,n})^X(x) = e^{u_n(x)} \int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x} ds$ , so Eq (7) can be written as  $e^{u_{n+1}(x)} = \mu(x)e^{u_n(x)}/(p_{n,n,n})^X(x)$ , which gives Eq (12). Hence, since  $(p_{n+1,n,n})^X = \mu$ ,

$$H(p_{n+1,n,n}|p_{n,n,n}) = \int \int p_{n+1,n,n}(s, x) \log \frac{\mu(x)}{(p_{n,n,n})^X(x)} dx ds = \int \mu(x) \log \frac{\mu(x)}{(p_{n,n,n})^X(x)} dx = H(\mu|(p_{n,n,n})^X),$$

which is Eq (13). ■

**Proposition 3.2** Suppose Assumption 2.2 holds. Then  $\|(p_{n,n,n})^X - \mu\|_{\text{TV}} \rightarrow 0$  as  $n \rightarrow \infty$ .<sup>5</sup>

**Proof.** By Lemma C.1, every step of the scheme is well defined and  $H(p^*|p_{n,n,n})$  and  $H(p^*|p_{n+1,n,n})$  are finite for every  $n \geq 0$ . Adding Eqs (C-1) and (C-2), the term  $H(p^*|p_{n+1,n,n})$  cancels and we obtain, for every  $n \geq 0$ ,

$$H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n}) = H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1}). \quad (14)$$

Now sum Eq (14) over  $n = 0, 1, \dots, N-1$ . On the right-hand side the sum telescopes, since the second term for index  $n$  cancels the first term for index  $n+1$ , leaving only the first term for  $n = 0$  and the second term for  $n = N-1$ :

$$\begin{aligned} \sum_{n=0}^{N-1} (H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})) &= \sum_{n=0}^{N-1} (H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n+1,n+1})) \\ &= H(p^*|p_{0,0,0}) - H(p^*|p_{N,N,N}) \\ &= H(p^*|R) - H(p^*|p_{N,N,N}) \leq H(p^*|R), \end{aligned}$$

since  $p_{0,0,0} = R$  and entropy is non-negative. Letting  $N \rightarrow \infty$  gives

$$\sum_{n=0}^{\infty} (H(p_{n+1,n,n}|p_{n,n,n}) + H(p_{n+1,n+1,n+1}|p_{n+1,n,n})) \leq H(p^*|R) \quad (15)$$

so in particular,  $H(p_{n+1,n,n}|p_{n,n,n}) \rightarrow 0$ . By Lemma 3.1,  $H(p_{n+1,n,n}|p_{n,n,n}) = H(\mu|(p_{n,n,n})^X)$ . Therefore  $H(\mu|(p_{n,n,n})^X) \rightarrow 0$  and Pinsker's inequality gives  $\|(p_{n,n,n})^X - \mu\|_{\text{TV}} \rightarrow 0$ . ■

### 3.1 Rate of convergence

**Proposition 3.3** Under the assumptions of Proposition 3.2, for every  $n \geq 1$ ,  $H((p_{n,n,n})^X|\mu) \leq \frac{H(p^*|R)}{n}$ . Hence  $\|(p_{n,n,n})^X - \mu\|_{L^1} \leq \sqrt{\frac{2H(p^*|R)}{n}}$ , and  $(p_{n,n,n})^X \rightarrow \mu$  in total variation at rate  $O(n^{-1/2})$ .

**Proof.** Write  $q_n = p_{n,n,n}$ . Since  $(p_{n+1,n,n})^X = \mu$ , the Data Processing inequality (DPI, cf. Lemma 1.6 in [Nutz22]) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}). \quad (16)$$

From Eq (12)  $p_{n+1,n,n}(s, x)/q_n(s, x) = \mu(x)/q_n^X(x)$  for  $n \geq 1$ , so  $q_n(s, x)/p_{n+1,n,n}(s, x) = q_n^X(x)/\mu(x)$   $q_n$ -almost everywhere, since for  $n \geq 1$  the density  $q_n^X$  is supported in  $(0, K)$ , where  $\mu > 0$  almost everywhere. Hence

$$H(q_n|p_{n+1,n,n}) = \int \log \frac{q_n(s, x)}{p_{n+1,n,n}(s, x)} q_n(s, x) ds dx = \int \log \frac{q_n^X(x)}{\mu(x)} q_n(s, x) ds dx = \int \log \frac{q_n^X(x)}{\mu(x)} q_n^X(x) dx = H(q_n^X|\mu). \quad (17)$$

In particular, by Eq (16) with  $n-1$  in place of  $n$  and Eq (15),  $H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu) \leq H(q_n|p_{n,n-1,n-1}) \leq H(p^*|R) < \infty$  for  $n \geq 1$ .

For  $n \geq 1$ , both  $q_n$  and  $q_{n+1}$  have  $S$ -marginal  $\nu$  and conditional mean  $b(s)$  by Lemma C.1. Moreover, as shown in the proof of Lemma C.1,  $q = p_{n+1,n,n}$  satisfies the hypotheses of Lemma B.1, with  $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$  and with  $q' = p_{n+1,n+1,n+1} = q_{n+1}$ . Hence Lemma B.1, with  $p = q_n$ , gives

$$H(q_n|p_{n+1,n,n}) = H(q_n|q_{n+1}) + H(q_{n+1}|p_{n+1,n,n}). \quad (18)$$

<sup>5</sup>see section 1.1 in [Nutz22] for the definition of the total variation distance  $\|\cdot\|_{\text{TV}}$  here

By Eq (18),  $H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n})$ . Combining this with Eqs (16) and (17) gives

$$H(q_{n+1}^X|\mu) \leq H(q_{n+1}|p_{n+1,n,n}) \leq H(q_n|p_{n+1,n,n}) = H(q_n^X|\mu).$$

On the other hand, Eqs (16) and (15) give

$$\sum_{n=1}^{\infty} H(q_n^X|\mu) \leq \sum_{n=0}^{\infty} H(q_{n+1}|p_{n+1,n,n}) \leq H(p^*|R).$$

Since  $H(q_n^X|\mu)$  is non-increasing and non-negative,  $nH(q_n^X|\mu) \leq \sum_{k=1}^n H(q_k^X|\mu) \leq H(p^*|R)$ , which proves the first claim. Finally, Pinsker's inequality gives  $\|q_n^X - \mu\|_{L^1} \leq \sqrt{2H(q_n^X|\mu)} \leq \sqrt{\frac{2H(p^*|R)}{n}}$ . ■

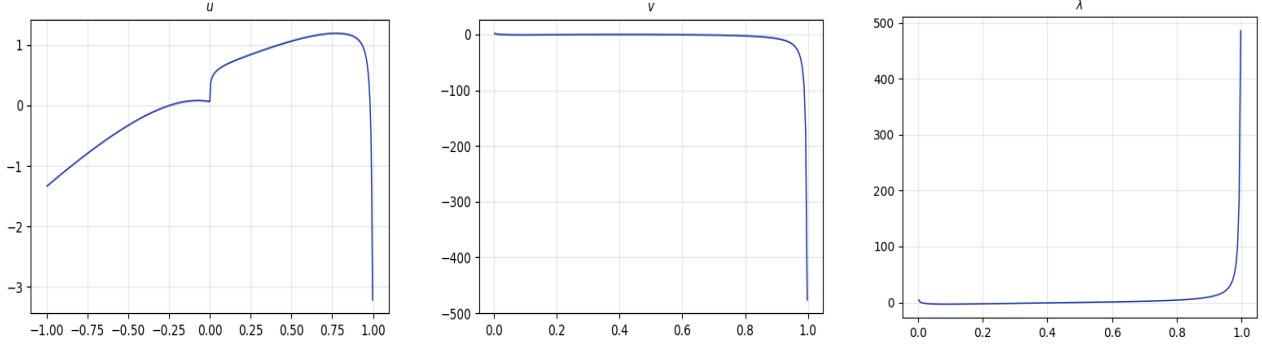


Figure 1: Here we have plotted  $u$ ,  $v$ ,  $\lambda$  for a target law  $\nu$  which is a 60/40 mixture of the Azéma-Yor and Perkins solutions respectively (and hence admissible) with  $\mu \sim U[-1, 1]$ , using 400 (equidistant) steps for  $X$  direction and 200 for  $S$ .

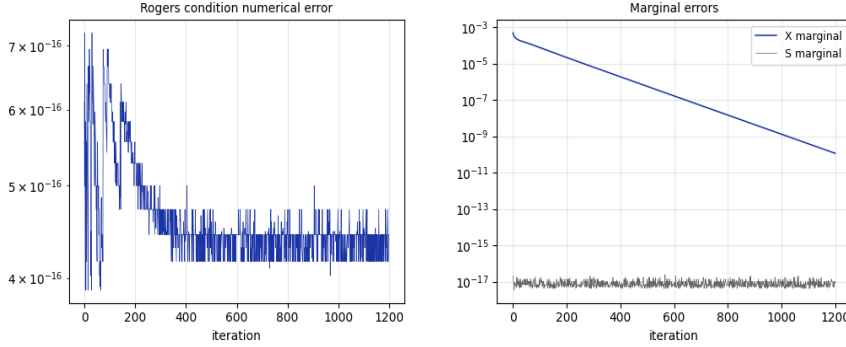


Figure 2: Here we have plotted the error for the Rogers condition and the  $\mu$ -marginal condition for the plots above.

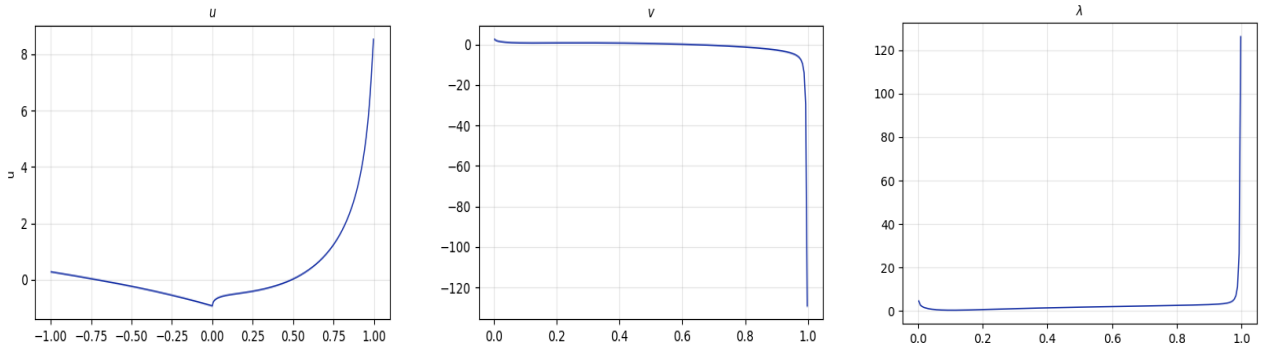


Figure 3: Here we have plotted  $u$ ,  $v$ ,  $\lambda$  for a target law  $\nu$  with cdf  $\nu([0, s]) = s^p$  with  $p = 3/4$ , again with  $\mu \sim U[-1, 1]$ .

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## A Proof of Lemma 2.3

**Proof.** Fix an  $s$  for which the hypothesis of the lemma holds, i.e. for which  $Z_s(\lambda) \in (0, \infty)$  for some finite  $\lambda$ , where  $Z_s(\lambda) = \int_0^s R(s, x)e^{u(x)+\lambda x}dx$ ; by assumption this is the case for  $\nu$ -almost every  $s$ . Suppose that, for some finite  $\lambda_2$ ,

$$0 < Z_s(\lambda_2) = \int_0^s R(s, x)e^{u(x)+\lambda_2 x}dx < \infty.$$

Then, for any other finite  $\lambda_1$ ,

$$Z_s(\lambda_1) = \int_0^s e^{(\lambda_1-\lambda_2)x}R(s, x)e^{u(x)+\lambda_2 x}dx.$$

Since  $x \in (0, s)$ , the factor  $e^{(\lambda_1-\lambda_2)x}$  is bounded above and below by strictly positive constants. Hence (by comparison with the previous expression)  $0 < Z_s(\lambda_1) < \infty$ . Therefore  $Z_s(\lambda)$  is positive and finite for every finite  $\lambda$ . Since  $x \in (0, s)$ , differentiation under the integral sign is justified locally in  $\lambda$  by dominated convergence. Moreover

$$\frac{d}{d\lambda} \log Z_s(\lambda) = \frac{\int_0^s x R(s, x)e^{u(x)+\lambda x}dx}{\int_0^s R(s, x)e^{u(x)+\lambda x}dx} =: m_s(\lambda),$$

which is the mean of  $x$  under the probability density  $R(s, x)e^{u(x)+\lambda x}/Z_s(\lambda)$  on  $(0, s)$ , and (using the quotient rule) we can easily check that  $m'_s(\lambda)$  is the corresponding variance,

$$m'_s(\lambda) = \int_0^s (x - m_s(\lambda))^2 \frac{R(s, x)e^{u(x)+\lambda x}}{Z_s(\lambda)} dx.$$

By Assumption 2.2,  $R(s, x) > 0$  for Lebesgue almost every  $x \in (0, s)$ . Therefore, since  $u(x)$  is finite Lebesgue almost everywhere, we have  $\frac{R(s, x)e^{u(x)+\lambda x}}{Z_s(\lambda)} > 0$  for Lebesgue almost every  $x \in (0, s)$ . The integrand is strictly positive for Lebesgue almost every  $x \in (0, s)$  except the single point  $x = m_s(\lambda)$ . Therefore

$$m'_s(\lambda) > 0$$

for every finite  $\lambda$ . Thus  $m_s$  is continuous and strictly increasing. Moreover, since  $R(s, x)e^{u(x)} > 0$  for Lebesgue almost every  $x \in (0, s)$ , for every  $\varepsilon \in (0, s)$  we have  $\int_0^\varepsilon R(s, x)e^{u(x)}dx > 0$  and  $\int_{s-\varepsilon}^s R(s, x)e^{u(x)}dx > 0$ . Now fix  $\lambda > 0$ . Since

$$Z_s(\lambda) = \int_0^s R(s, x)e^{u(x)}e^{\lambda x}dx,$$

and the integrand is non-negative,  $Z_s(\lambda) \geq \int_{s-\varepsilon}^s R(s, x) e^{u(x)} e^{\lambda x} dx$ . For  $x \in (s - \varepsilon, s)$ ,  $e^{\lambda x} \geq e^{\lambda(s-\varepsilon)}$ , so

$$Z_s(\lambda) \geq e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s, x) e^{u(x)} dx.$$

Also, since  $x < s$ ,  $e^{\lambda x} \leq e^{\lambda s}$ , and hence  $Z_s(\lambda) \leq e^{\lambda s} \int_0^s R(s, x) e^{u(x)} dx = e^{\lambda s} Z_s(0)$ . Therefore

$$e^{\lambda(s-\varepsilon)} \int_{s-\varepsilon}^s R(s, x) e^{u(x)} dx \leq Z_s(\lambda) \leq e^{\lambda s} Z_s(0).$$

Taking logarithms and dividing by  $\lambda$  gives

$$s - \varepsilon + o(1) \leq \frac{\log Z_s(\lambda)}{\lambda} \leq s + o(1)$$

as  $\lambda \rightarrow \infty$ . Since  $\varepsilon > 0$  is arbitrary,  $\lim_{\lambda \rightarrow \infty} \log Z_s(\lambda)/\lambda = s$ . Similarly, using  $\int_0^\varepsilon R(s, x) e^{u(x)} dx > 0$  and  $e^{\lambda x} \geq e^{\lambda \varepsilon}$  for  $x \in (0, \varepsilon)$  and  $\lambda < 0$ ,  $\lim_{\lambda \rightarrow -\infty} \frac{\log Z_s(\lambda)}{\lambda} = 0$ . We now use the following elementary fact: for a differentiable convex function  $f$  on  $\mathbb{R}$ , the limits  $\lim_{\lambda \rightarrow \pm\infty} f'(\lambda)$  exist in  $[-\infty, \infty]$  and  $\lim_{\lambda \rightarrow \pm\infty} f(\lambda)/\lambda = \lim_{\lambda \rightarrow \pm\infty} f'(\lambda)$ . The function  $f = \log Z_s$  is differentiable with  $f' = m_s$ , and convex because  $f'' = m'_s \geq 0$  by the variance formula above. Applying this to  $f = \log Z_s$  and combining it with the two limits just computed, we obtain

$$\lim_{\lambda \rightarrow -\infty} m_s(\lambda) = 0, \quad \lim_{\lambda \rightarrow \infty} m_s(\lambda) = s.$$

Recall from Eq (10) that  $0 < b(s) < s$ , so (by the intermediate value theorem) there exists a unique finite  $\lambda(s)$  satisfying

$$m_s(\lambda(s)) = b(s).$$

Since  $m_s(\lambda)$  is measurable in  $s$  for every fixed  $\lambda$  and continuous and strictly increasing in  $\lambda$  for  $\nu$ -almost every  $s$ , the unique root  $\lambda(s)$  admits a measurable version.

The final assertion of the lemma was established at the start of the proof, where  $Z_s(\lambda)$  was shown to be positive and finite for every finite  $\lambda$ . ■

## B The entropy projection onto the $S$ -marginal and Rogers constraints.

The following lemma describes what happens when the  $(\lambda, \nu)$ -update is applied to a probability density  $q$ : the resulting density  $q'$  satisfies two of the three constraints, namely the  $S$ -marginal and the local Rogers conditions, and for any other density  $p$  satisfying these two constraints the relative entropies of  $p$  and  $q'$  with respect to  $q$  are related by the Pythagorean identity  $H(p|q) = H(p|q') + H(q'|q)$ . In particular  $q'$  is the density closest to  $q$  in relative entropy among those satisfying the two constraints.

**Lemma B.1** *Let  $D = \{(s, x) : s \geq X_0, 0 < x \leq s\}$ , and let  $q$  be a probability density on  $D$  whose  $S$ -marginal  $q^S$  is positive  $\nu$ -almost everywhere. For a given measurable function  $\lambda$  which is finite  $\nu$ -almost everywhere, define*

$$q_\lambda(x|s) = \frac{e^{\lambda(s)x} q(s, x)}{\int_0^s e^{\lambda(s)y} q(s, y) dy}, \quad x \in (0, s], \text{ for every } s \text{ with } q^S(s) > 0,$$

and suppose that  $\lambda$  is such that

$$\int_0^s x q_\lambda(x|s) dx = b(s) \quad \text{for } \nu\text{-almost every } s. \quad (\text{B-1})$$

Set  $q'(s, x) = \nu(s) q_\lambda(x|s)$  for  $s$  with  $q^S(s) > 0$  and  $q'(s, x) = 0$  otherwise, i.e.  $q'$  has  $S$ -marginal  $\nu$  and conditional density  $q_\lambda(\cdot|s)$  given  $S = s$ .

Then  $q'$  is a probability density on  $D$ , and for any probability density  $p$  on  $D$  with  $S$ -marginal  $\nu$ , conditional mean  $\int_0^s x p(x|s) dx = b(s)$  for  $\nu$ -almost every  $s$ , and  $H(p|q) < \infty$ , we have  $H(p|q) = H(p|q') + H(q'|q)$ .

**Proof. Step 1: properties of  $q'$ .** For  $s$  with  $\nu(s) > 0$ , integrating the definition of  $q'$  over  $x$  gives  $\int_0^s q'(s, x) dx = \nu(s)$ , and for  $s$  with  $\nu(s) = 0$  both sides vanish; hence  $q'$  has  $S$ -marginal  $\nu$  and is a probability density on  $D$ . Its conditional density given  $S = s$  is  $q'(x|s) = e^{\lambda(s)x} q(s, x) / \int_0^s e^{\lambda(s)y} q(s, y) dy$ , so the defining equation for  $\lambda$  says precisely that  $\int_0^s x q'(x|s) dx = b(s)$  for  $\nu$ -almost every  $s$ :  $q'$  has the same conditional mean as  $p$ . Moreover, dividing by  $q(x|s) = q(s, x)/q^S(s)$ ,

$$\frac{q'(x|s)}{q(x|s)} = \frac{q^S(s) e^{\lambda(s)x}}{\int_0^s e^{\lambda(s)y} q(s, y) dy} \quad \text{for } \nu\text{-almost every } s, \quad (\text{B-2})$$

whose right-hand side is, for  $\nu$ -almost every  $s$ , positive and finite for every  $x \in (0, s)$ , since  $q^S(s) > 0$ ,  $\lambda(s)$  is finite and  $q(s, \cdot)$  is integrable. Hence  $q'(\cdot|s)$  and  $q(\cdot|s)$  vanish on the same set, i.e. they are mutually absolutely continuous, which we write  $q(\cdot|s) \sim q'(\cdot|s)$ . Taking logarithms, and noting that neither  $q^S(s)$  nor the integral in the denominator depends on  $x$ ,

$$\log \frac{q'(x|s)}{q(x|s)} = \lambda(s)x + c(s), \quad c(s) := \log q^S(s) - \log \int_0^s e^{\lambda(s)y} q(s, y) dy, \quad (\text{B-3})$$

so the log-density ratio is an affine function of  $x$ .

**Step 2: decomposing relative entropy along  $S$ .** Let  $P$  and  $Q$  be probability densities on  $D$  with  $P \ll Q$ , and write  $P(s, x) = P^S(s)P(x|s)$  and  $Q(s, x) = Q^S(s)Q(x|s)$ . Then  $\log \frac{P(s, x)}{Q(s, x)} = \log \frac{P^S(s)}{Q^S(s)} + \log \frac{P(x|s)}{Q(x|s)}$  for  $P$ -almost every  $(s, x)$ , and integrating against  $P$  gives

$$H(P|Q) = H(P^S|Q^S) + \int P^S(s) H(P(\cdot|s)|Q(\cdot|s)) ds. \quad (\text{B-4})$$

**Step 3: the identity fibrewise.** Applying Eq (B-4) with  $P = p$ ,  $Q = q$  and using  $H(p|q) < \infty$ , both terms on the right-hand side are finite; in particular  $H(p(\cdot|s)|q(\cdot|s)) < \infty$  for  $\nu$ -almost every  $s$ . Fix such an  $s$ . Since  $q'(\cdot|s)$  is equivalent to  $q(\cdot|s)$  by Step 1, Lemma 1.4(b) in [Nutz22] gives

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + \int_0^s \log \frac{q'(x|s)}{q(x|s)} p(x|s) dx.$$

By Eq (B-3), for any probability density  $\rho$  on  $(0, s)$  the integral  $\int_0^s \log \frac{q'(x|s)}{q(x|s)} \rho(x) dx = \lambda(s) \int_0^s x \rho(x) dx + c(s)$  depends on  $\rho$  only through its mean. Since  $p(\cdot|s)$  and  $q'(\cdot|s)$  both have mean  $b(s)$  (by hypothesis and by Step 1 respectively), the last integral above is unchanged if  $p(x|s)$  is replaced by  $q'(x|s)$ , and equals  $H(q'(\cdot|s)|q(\cdot|s))$ . Therefore

$$H(p(\cdot|s)|q(\cdot|s)) = H(p(\cdot|s)|q'(\cdot|s)) + H(q'(\cdot|s)|q(\cdot|s)) \quad \text{for } \nu\text{-almost every } s. \quad (\text{B-5})$$

**Step 4: integrating over  $s$ .** Apply Eq (B-4) to the three pairs  $(p, q)$ ,  $(p, q')$  and  $(q', q)$ ; the absolute continuity required holds since  $p \ll q$  by hypothesis,  $q' \ll q$  by Step 1, and  $p \ll q'$  because  $p(\cdot|s) \ll q(\cdot|s)$  (from  $p \ll q$ ) and  $q(\cdot|s) \sim q'(\cdot|s)$  (Step 1, after Eq (B-2)) for  $\nu$ -almost every  $s$ , and both  $p$  and  $q'$  have  $S$ -marginal  $\nu$ .

Multiplying Eq (B-5) by  $\nu(s)$  and integrating over  $s$  gives

$$\int \nu(s) H(p(\cdot|s)|q(\cdot|s)) ds = \int \nu(s) H(p(\cdot|s)|q'(\cdot|s)) ds + \int \nu(s) H(q'(\cdot|s)|q(\cdot|s)) ds. \quad (\text{B-6})$$

We now express each of the three integrals in Eq (B-6) via Eq (B-4), whose first term on the right is the relative entropy of the  $S$ -marginals. Here  $p^S = \nu$  by hypothesis and  $q'^S = \nu$  by Step 1, while  $q^S$  is the  $S$ -marginal of  $q$ . Recall (B-4) states that  $H(P|Q) = H(P^S|Q^S) + \int P^S(s) H(P(\cdot|s)|Q(\cdot|s)) ds$ . Then taking  $(P, Q) = (p, q)$ ,  $(p, q')$  and  $(q', q)$  respectively gives

$$H(p|q) = H(\nu|q^S) + \int \nu(s) H(p(\cdot|s)|q(\cdot|s)) ds, \quad (\text{B-7})$$

$$H(p|q') = \int \nu(s) H(p(\cdot|s)|q'(\cdot|s)) ds, \quad (\text{B-8})$$

$$H(q'|q) = H(\nu|q^S) + \int \nu(s) H(q'(\cdot|s)|q(\cdot|s)) ds. \quad (\text{B-9})$$

Substituting Eqs (B-7), (B-8) and (B-9) into Eq (B-6), i.e. replacing the three integrals there by  $H(p|q) - H(\nu|q^S)$ ,  $H(p|q')$  and  $H(q'|q) - H(\nu|q^S)$  respectively, the two terms  $H(\nu|q^S)$  cancel and we obtain  $H(p|q) = H(p|q') + H(q'|q)$ .  $\blacksquare$

## C Well-definedness of the Sinkhorn scheme

For  $n \geq 1$ , we regard  $u_n$  and  $v_n$  as extended-real-valued functions, with  $u_n(x) = -\infty$  for  $x \notin (0, K)$  and  $v_n(s) = -\infty$  for  $s \notin (X_0, K)$ . The multiplier  $\lambda_n$  is only relevant  $\nu$ -almost everywhere, and we set  $\lambda_n(s) = 0$  whenever  $\nu(s) = 0$ . Thus  $p_{n,n,n}(s, x) = R(s, x) e^{u_n(x) + v_n(s) + \lambda_n(s)x}$  vanishes whenever  $x \notin (0, K)$  or  $s \notin (X_0, K)$ .

**Lemma C.1** *Suppose Assumption 2.2 holds and set  $u_0 = v_0 = \lambda_0 = 0$ . Then every step of the Sinkhorn scheme is well defined. For every  $n \geq 0$ ,  $H(p^*|p_{n,n,n}) < \infty$ ,  $H(p^*|p_{n+1,n,n}) < \infty$  and*

$$H(p^*|p_{n,n,n}) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|p_{n,n,n}), \quad (\text{C-1})$$

$$H(p^*|p_{n+1,n,n}) - H(p^*|p_{n+1,n+1,n+1}) = H(p_{n+1,n+1,n+1}|p_{n+1,n,n}). \quad (\text{C-2})$$

The functions  $u_n$  are finite Lebesgue almost everywhere on  $(0, K)$ , and  $\lambda_n$  and  $v_n$  are finite  $\nu$ -almost everywhere. Every completed iterate  $p_{n,n,n}$  with  $n \geq 1$  has  $S$ -marginal  $\nu$  and satisfies the local Rogers condition.

**Proof.** We proceed by induction. Since  $p_{0,0,0} = R$ , Assumption 2.2 gives  $H(p^*|p_{0,0,0}) < \infty$ . Also  $u_0 = 0$ ,  $\lambda_0 = 0$  and  $v_0 = 0$  are finite, so the finiteness assertions hold at  $n = 0$ . As the inductive hypothesis, we assume that  $H(p^*|p_{n,n,n}) < \infty$ ,  $u_n$  is finite Lebesgue almost everywhere on  $(0, K)$ , and that  $\lambda_n$  and  $v_n$  are finite  $\nu$ -almost everywhere.

Write  $q_n = p_{n,n,n}$  and let  $q_n^X$  denote its  $X$ -marginal. Since the  $X$ -marginals of  $p^*$  and  $q_n$  are  $\mu$  and  $q_n^X$ , respectively, the DPI gives

$$H(\mu|q_n^X) \leq H(p^*|q_n) < \infty. \quad (\text{C-3})$$

Since  $H(\mu|q_n^X) < \infty$ , we have  $\mu \ll q_n^X$ , so the set  $\{x \in (0, K) : q_n^X(x) = 0\}$  is  $\mu$ -null; as  $\mu > 0$  Lebesgue almost everywhere on  $(0, K)$  by Assumption 2.2, this set is Lebesgue-null, i.e.  $q_n^X(x) > 0$  for Lebesgue almost every  $x \in (0, K)$ . Moreover, since  $q_n^X$  is a probability density,  $q_n^X(x) < \infty$  for Lebesgue almost every  $x \in (0, K)$ .

Recall the  $u$ -update, Eq (7):  $u_{n+1}(x) = \log \mu(x) - \log(\int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x} ds)$ . Integrating  $q_n(s, x) = R(s, x)e^{u_n(x)+v_n(s)+\lambda_n(s)x}$  over  $s$  gives  $q_n^X(x) = e^{u_n(x)} \int_x^\infty R(s, x)e^{v_n(s)+\lambda_n(s)x} ds$ , so the integral inside the logarithm above equals  $e^{-u_n(x)} q_n^X(x)$ , and therefore

$$u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$$

for Lebesgue almost every  $x \in (0, K)$ . Hence  $p_{n+1,n,n}(s, x)/q_n(s, x) = e^{u_{n+1}(x)-u_n(x)} = \mu(x)/q_n^X(x)$ , so  $p_{n+1,n,n} \ll q_n$ , and  $p^* \ll q_n$  since  $H(p^*|q_n) < \infty$  by the inductive hypothesis.

Lemma 1.4(b) in [Nutz22] gives

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = \mathbb{E}^{p^*}(\log \frac{p_{n+1,n,n}(S, X)}{q_n(S, X)}) = \mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}).$$

Since the  $X$ -marginal of  $p^*$  is  $\mu$ ,  $\mathbb{E}^{p^*}(\log \frac{\mu(X)}{q_n^X(X)}) = H(\mu|q_n^X)$ , and therefore  $H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(\mu|q_n^X)$ . Since  $H(p^*|q_n) < \infty$  and  $H(\mu|q_n^X) < \infty$ , it follows that  $H(p^*|p_{n+1,n,n}) < \infty$ .

Moreover, by Eq (13),  $H(p_{n+1,n,n}|q_n) = H(\mu|q_n^X)$ . Hence

$$H(p^*|q_n) - H(p^*|p_{n+1,n,n}) = H(p_{n+1,n,n}|q_n)$$

which is Eq (C-1). By Eq (13),  $H(p_{n+1,n,n}|q_n) = H(\mu|q_n^X)$  and by (C-3) this quantity is finite.

Since  $\mu(x) > 0$  and  $0 < q_n^X(x) < \infty$  for Lebesgue almost every  $x \in (0, K)$  (shown above), and  $u_n$  is finite Lebesgue almost everywhere on  $(0, K)$  by the inductive hypothesis, the identity  $u_{n+1}(x) = \log \mu(x) - \log q_n^X(x) + u_n(x)$  shows that  $u_{n+1}$  is finite Lebesgue almost everywhere on  $(0, K)$ .

We now apply Lemma 2.3 directly to  $u_{n+1}$ . Since  $p_{n+1,n,n}(s, x) = R(s, x)e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$ , its  $S$ -marginal is

$$(p_{n+1,n,n})^S(s) = e^{v_n(s)} \int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_n(s)x} dx.$$

Since  $p_{n+1,n,n}$  is a probability density,  $(p_{n+1,n,n})^S$  is integrable, hence finite for Lebesgue almost every  $s$ . As shown above,  $H(p^*|p_{n+1,n,n}) < \infty$ . Since the  $S$ -marginal of  $p^*$  is  $\nu$ , the DPI gives  $H(\nu|(p_{n+1,n,n})^S) \leq H(p^*|p_{n+1,n,n}) < \infty$ , hence  $\nu \ll (p_{n+1,n,n})^S$ , i.e.  $(p_{n+1,n,n})^S(s) > 0$  for  $\nu$ -almost every  $s$ .

We have already shown that  $0 < (p_{n+1,n,n})^S(s) < \infty$  for  $\nu$ -almost every  $s$ . Moreover,  $v_n$  is finite  $\nu$ -almost everywhere by the inductive hypothesis, and hence  $0 < e^{-v_n(s)} < \infty$  for  $\nu$ -almost every  $s$ . Therefore

$$0 < \int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_n(s)x} dx < \infty \quad (\text{C-4})$$

for  $\nu$ -almost every  $s$ . Hence the hypotheses of Lemma 2.3 hold, so the  $\lambda$ -update is well defined and there exists a unique finite measurable function  $\lambda_{n+1}(s)$  satisfying

$$\frac{\int_0^s x R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx}{\int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx} = b(s).$$

By Assumption 2.2 and the fact that  $\nu$  is a probability density,  $0 < \nu(s) < \infty$  for  $\nu$ -almost every  $s$ . Now recall Eq (9):

$$v_{n+1}(s) = \log \nu(s) - \log(\int_0^s R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} dx).$$

Then  $\log \nu(s)$  is finite for  $\nu$ -almost every  $s$ , and the second logarithm in Eq (9) is finite for  $\nu$ -almost every  $s$  by Lemma 2.3 applied with  $u = u_{n+1}$  and  $\lambda = \lambda_{n+1}(s)$ , so the  $v$ -update is well defined and  $v_{n+1}$  is finite  $\nu$ -almost everywhere.

We now apply Lemma B.1 with  $q = p_{n+1,n,n}$  and  $\lambda(s) = \lambda_{n+1}(s) - \lambda_n(s)$ . Its hypotheses hold. First, the  $S$ -marginal of  $q$  is positive  $\nu$ -almost everywhere (shown above). Second, by definition  $q_\lambda(x|s) \propto e^{\lambda(s)x} q(s, x)$ ; with  $\lambda = \lambda_{n+1} - \lambda_n$  and  $q = p_{n+1,n,n} = R e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x}$  this is

$$e^{(\lambda_{n+1}(s)-\lambda_n(s))x} R(s, x)e^{u_{n+1}(x)+v_n(s)+\lambda_n(s)x} = e^{v_n(s)} R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x},$$

Hence  $q_\lambda(x|s) = R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} / \int_0^s R(s, y)e^{u_{n+1}(y)+\lambda_{n+1}(s)y}dy$ , whose mean  $\int_0^s xq_\lambda(x|s)dx$  is the left-hand side of Eq (8), hence equal to  $b(s)$ . Finally, by the definition of  $q'$  in Lemma B.1,

$$q'(s, x) = \nu(s) q_\lambda(x|s) = \frac{\nu(s)}{\int_0^s R(s, y)e^{u_{n+1}(y)+\lambda_{n+1}(s)y}dy} R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} = e^{v_{n+1}(s)} R(s, x)e^{u_{n+1}(x)+\lambda_{n+1}(s)x} = p_{n+1, n+1, n+1}$$

where the third equality is Eq (9).

$p^*$  has  $S$ -marginal  $\nu$ , conditional mean  $b$  and  $H(p^*|p_{n+1, n, n}) < \infty$  (shown above), so Lemma B.1 gives

$$H(p^*|p_{n+1, n, n}) = H(p^*|p_{n+1, n+1, n+1}) + H(p_{n+1, n+1, n+1}|p_{n+1, n, n}) \quad (\text{C-5})$$

which is Eq (C-2). Since the left-hand side is finite and both terms on the right are nonnegative,

$$H(p^*|p_{n+1, n+1, n+1}) < \infty.$$

Finally,  $(p_{n+1, n+1, n+1})^S = \nu$  by Lemma B.1, since  $p_{n+1, n+1, n+1}$  is the density  $q'$  there, and by Eq (8)  $\int_0^s xp_{n+1, n+1, n+1}(s, x)dx = b(s)\nu(s) = r(s)$ , so the completed iterate satisfies the local Rogers condition. This completes the induction. ■

## D Sketch proof of the Rogers' construction

For completeness we sketch the construction in Section 3 of [Rog93] of a continuous martingale with a prescribed joint law of  $(S_T, X_T)$ , in the case relevant here where the joint law has a density. It is convenient to work with  $Y = S - X$  and to write  $\mu(b, y)$  for the joint density of  $(S_T, Y_T)$  on  $[x_0, \infty) \times (0, \infty)$ , so that  $\mu(b, y) = \pi(b, b - y)$  in the notation of Theorem 1.1.

Let  $W$  be a Brownian motion started at  $x_0$  with running maximum  $\bar{W}_t = \max_{u \leq t} W_u$ , and let  $\{Z_b\}_{b \geq x_0}$  be a family of random variables, independent of  $W$  and of each other, with  $Z_b$  having density  $f(\cdot; b)$  on  $(0, \infty)$ . Define  $\tau = \inf\{t \geq 0 : \bar{W}_t - W_t \geq Z_{\bar{W}_t}\}$ . The rate of exclusions below the level  $y$  is  $\frac{1}{y}$ , so the excursions that reach  $Z_b$  arrive at rate  $\int_0^\infty y^{-1} f(y; b) dy$ , and let  $\tau$  is the time of the first such excursion. Then

$$\begin{aligned} \mathbb{P}(\bar{W}_\tau > b) &= \exp\left[-\int_{x_0}^b \int_0^\infty \frac{1}{y} f(y; s) dy ds\right], \\ \mathbb{P}(\bar{W}_\tau \in db, \bar{W}_\tau - W_\tau \in dy) &= \mathbb{P}(\bar{W}_\tau \geq b) \frac{1}{y} f(y; b) dy db. \end{aligned} \quad (\text{D-1})$$

Now let  $\mu(b, y)$  be the target density and  $\rho(b) = \int_b^\infty \int_0^\infty \mu(s, u) du ds = \mathbb{P}(S_T > b)$ . Matching (D-1) with  $\mu$  requires  $\mathbb{P}(\bar{W}_\tau \geq b) = \rho(b)$  and

$$f(y; b) = \frac{y\mu(b, y)}{\rho(b)} \quad (\text{D-2})$$

This is a legitimate choice only if  $f(\cdot; b)$  is a probability density in  $y$  for each  $b$ , i.e.

$$\int_0^\infty y\mu(b, y) dy = \rho(b) \quad \text{for almost every } b \geq x_0. \quad (\text{D-3})$$

Substituting  $\mu(b, y) = \pi(b, b - y)$ , the left-hand side of (D-3) is  $\int_{-\infty}^b (b - x)\pi(b, x) dx = b\nu(b) - \eta(b)$  and  $\rho(b) = \bar{F}_\nu(b)$ , so (D-3) is exactly the local Rogers condition (1).